Key features of parabolas

1

a Downwards

b Upwards

2 Upward

3 No, because the entire parabola will be above the x-axis. Also, there is no solution for the equation $x^2+6=0$.

4

a No

b Yes

c No

d Yes

5

a $x=0,4$

b $y=0$

c 4

6

a Concave up

b $y=9$

c $y=1$

d $x=-2$

e $x<-2$

7

a $(-2,0)$

b Vertex: $(-2,0)$

c

i False

ii True

iii True

iv True

8

a 2

b 1

9 Concave up

10

a $(-4,2)$

b 2

11 $x=3$

12

a No

b No

c Yes

13 Quadratic function A

15 $y = 25$

16

a $x = 14, 6$

b $x = 10$

c Concave down

d $y = 16$

17 The x-coordinate of the vertex of the parabola occurs at $x = \dfrac{-b}{2a}$. The y-coordinate of the vertex is found by substituting $x = \dfrac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x.

18 $x = \dfrac{1}{4}$

19 $x = -3, 2$

20

a False

b True

21 $x = \dfrac{-3 \pm \sqrt{93}}{6}$

Graphing quadratics

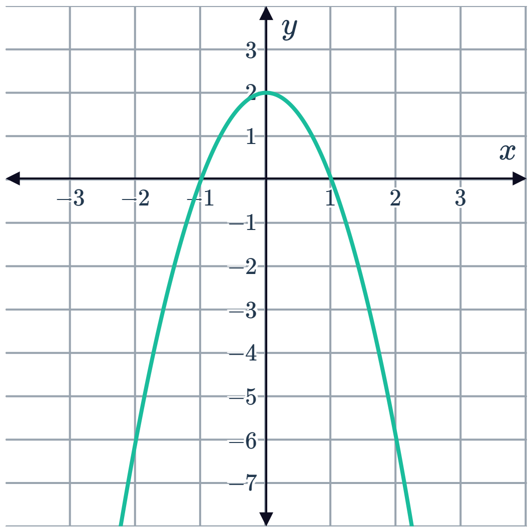
22

a Concave down

b More steep

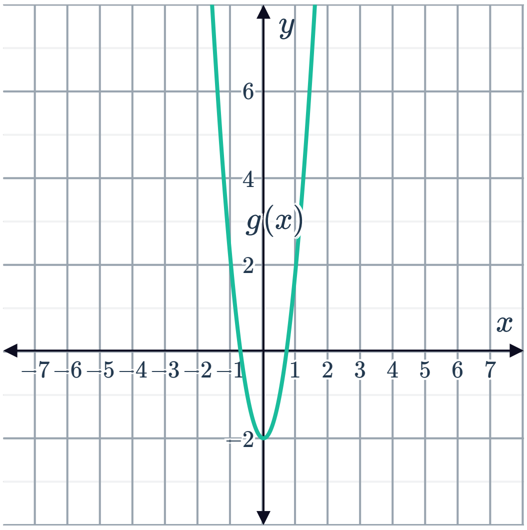
c $(0, 2)$

d

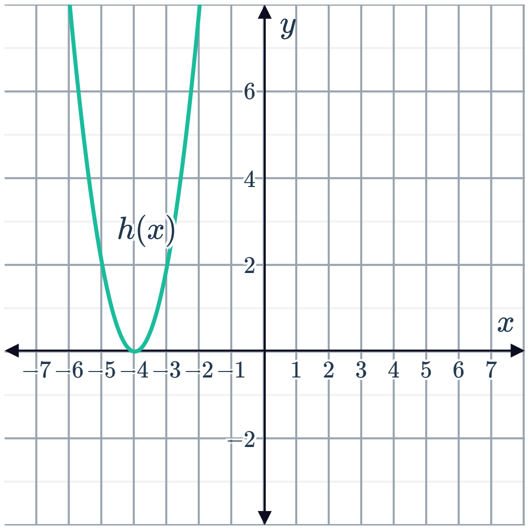


23

a



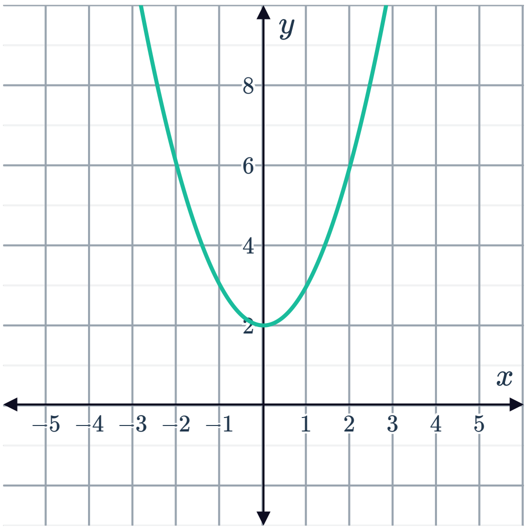
b



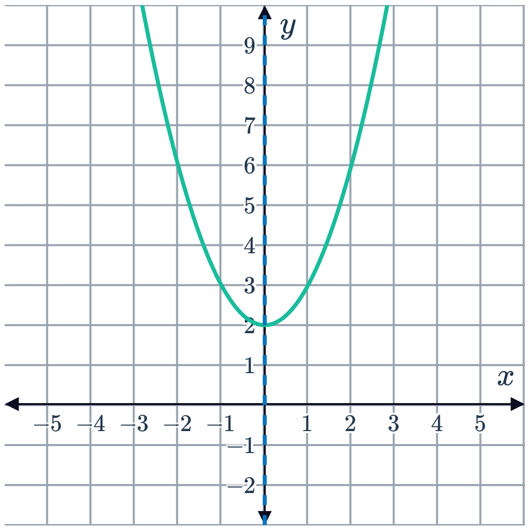
24 $y = -x^2 + 5$

25

a



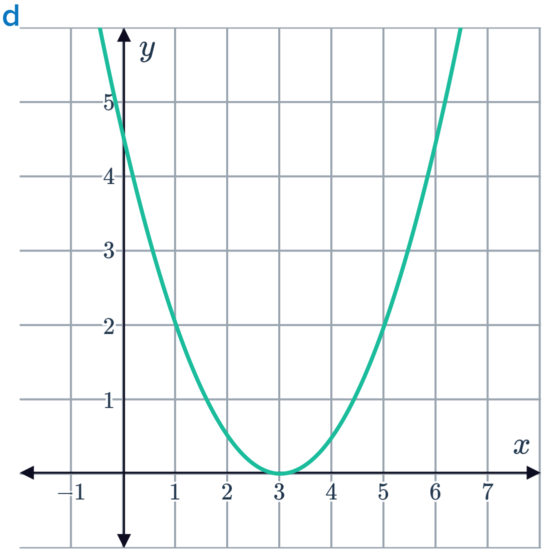
b



c $(0, 2)$

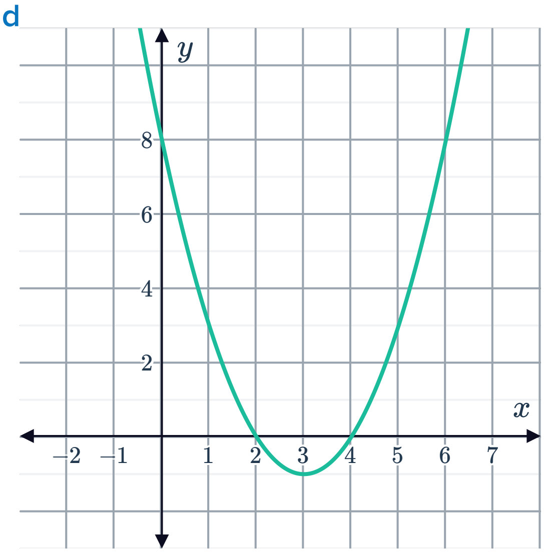
26

a Concave up **b** Less steep **c** $(3, 0)$ **d**



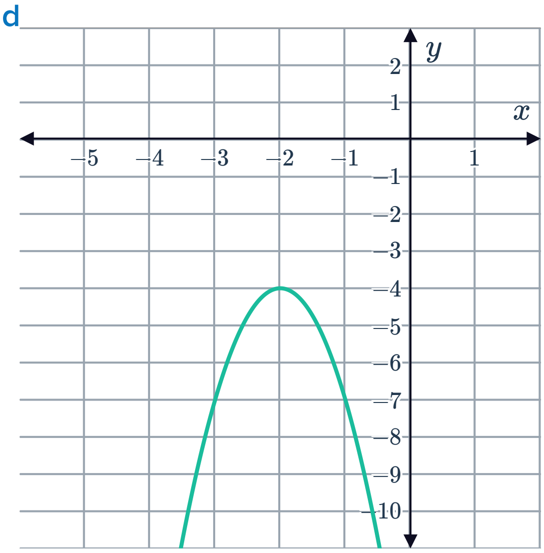
27

a $x = 4, 2$ **b** $y = 8$ **c** $(3, -1)$ **d**

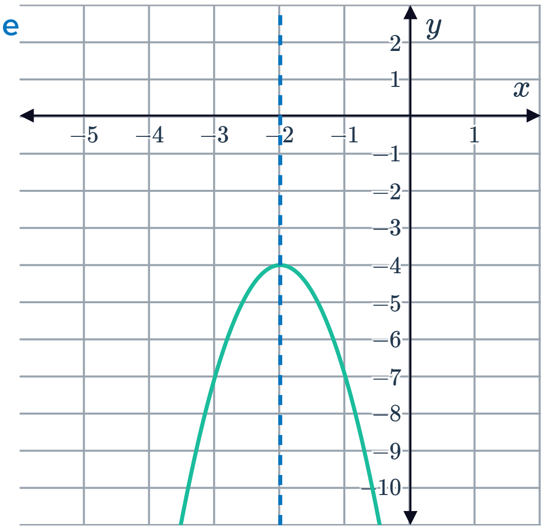


28

a $(-2, -4)$ **b** $x = -2$ **c** $y = -7$ **d**

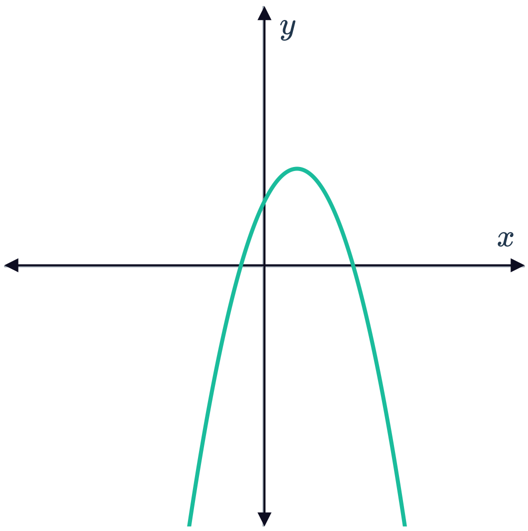


e

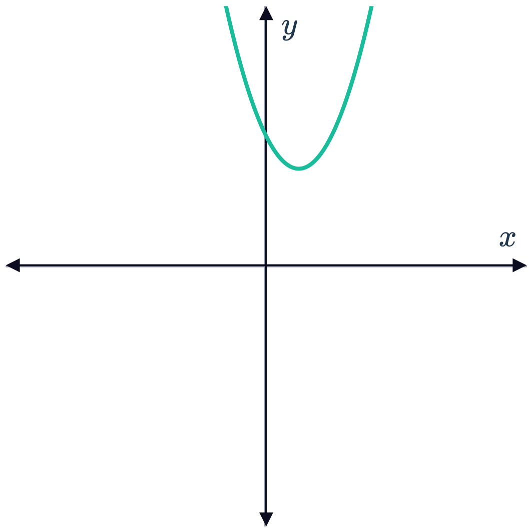


29

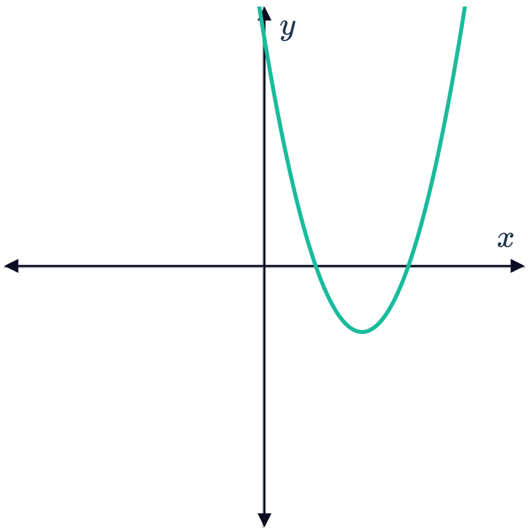
a



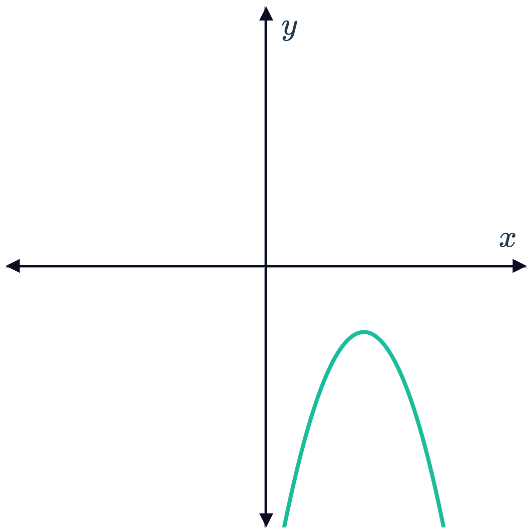
b



c



d



30

a 8

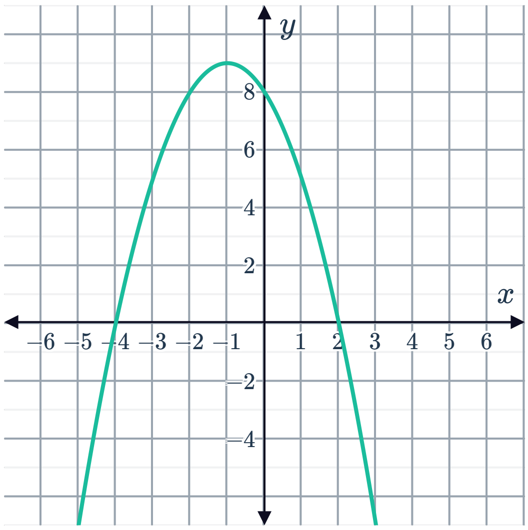
b $x = 2, -4$

c

x	-5	-3	-1	1	3
y	-7	5	9	5	-7

d $(-1, 9)$

e



31

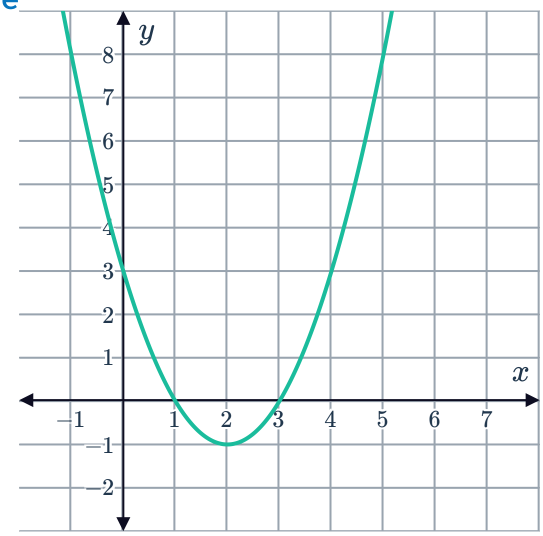
a $y = 3$

b $x = 3, 1$

c $x = 2$

d $(2, -1)$

e



32

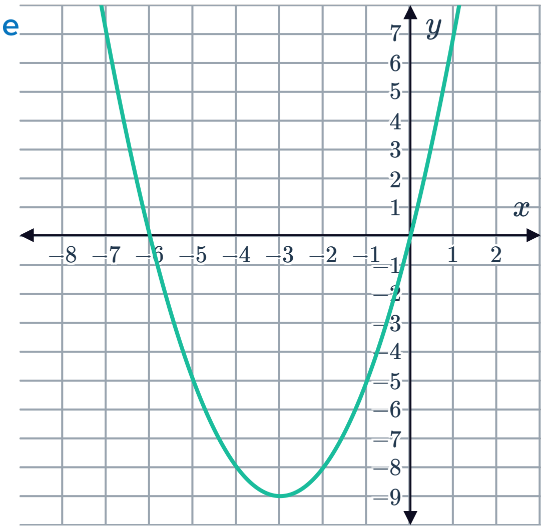
a $y = 0$

b $x = 0, -6$

c $x = -3$

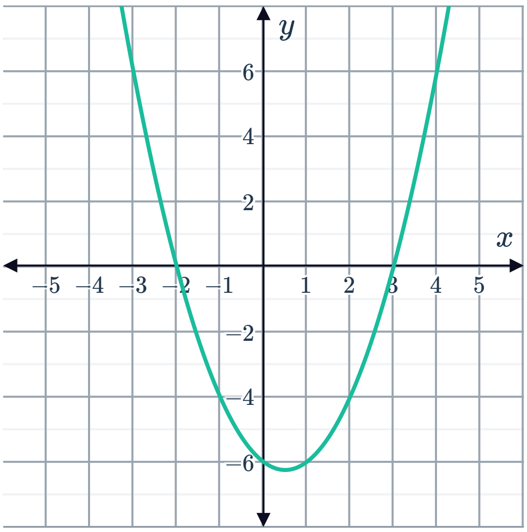
d $(-3, -9)$

e

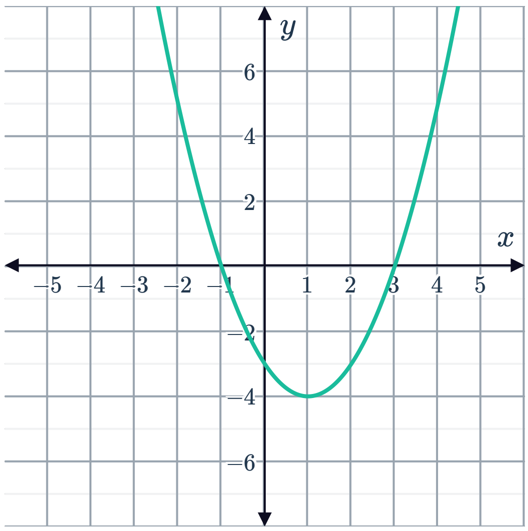


33

a

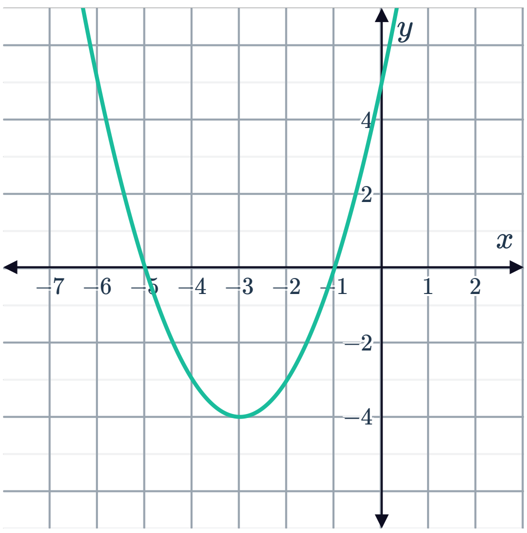


b

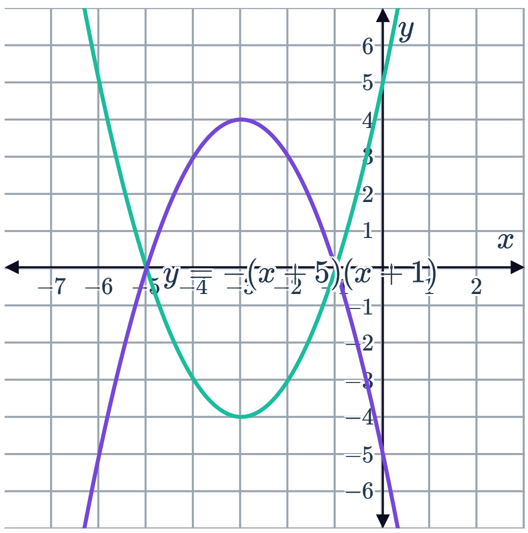


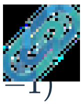
34

a



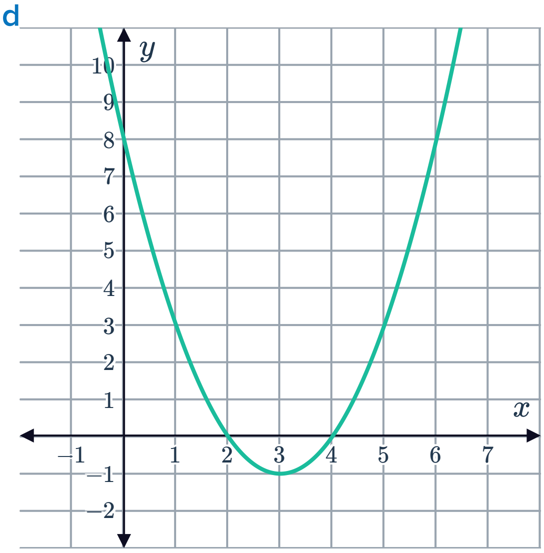
b





35

a $(x-2)(x-4)$ **b** $x=2,4$ **c** $(3,-1)$ **d**



36

a $y=(x+4)(x-3)$

b $x=-4,3$

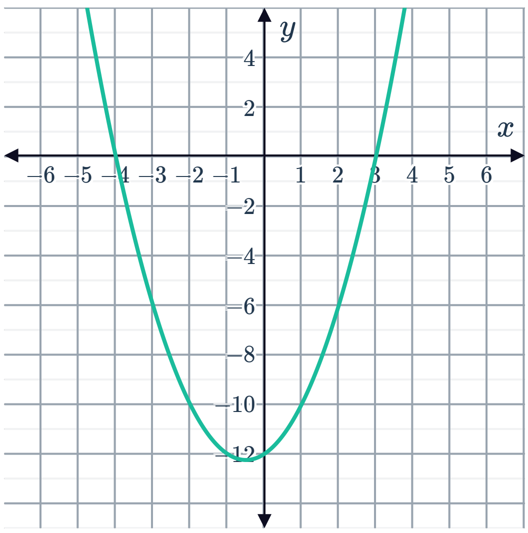
c $y=-12$

d $x=-\dfrac{1}{2}$

e $\left(-\dfrac{1}{2},-\dfrac{49}{4}\right)$

f Concave up

g

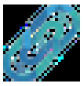


37

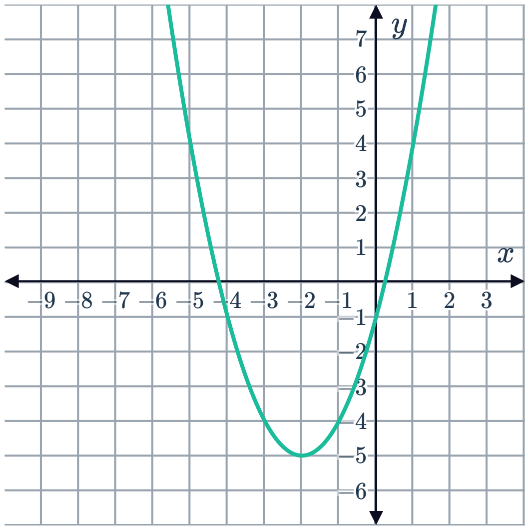
a $y=-1$

b $(-2,-5)$

c Concave up



d

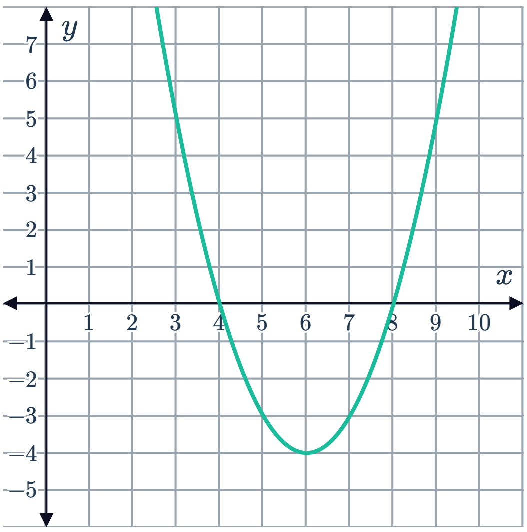


38

a $x = 8, 4$

b $(6, -4)$

c

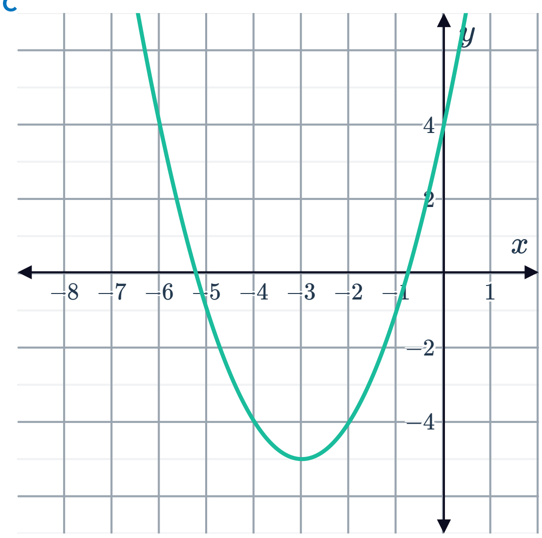


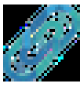
39

a $x = -3$

b -5

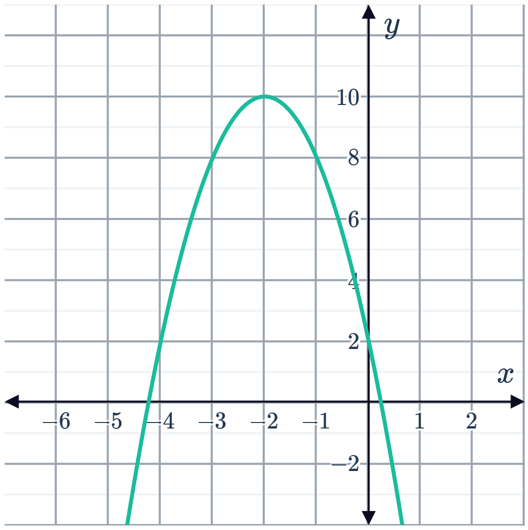
c





40 **a** $(-2, 10)$

b

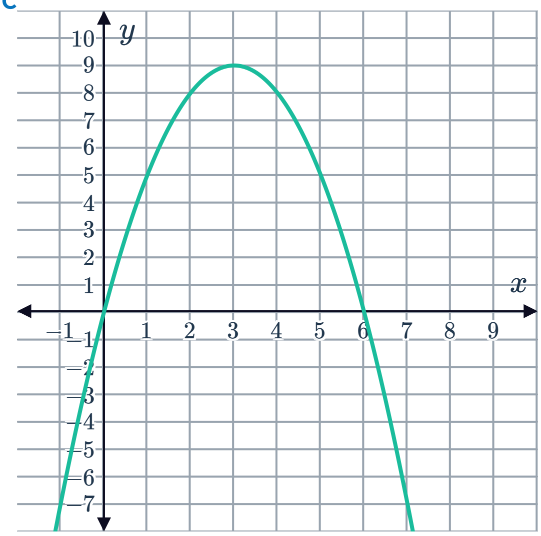


41

a $x = 0, 6$

b $(3, 9)$

c



42

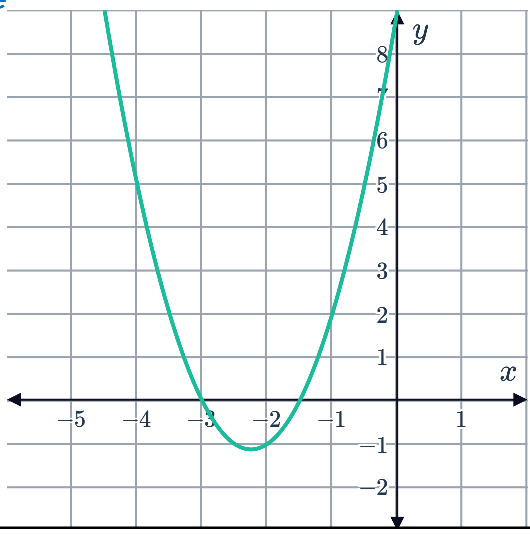
a $x = -\frac{3}{2}, -3$

b $y = 9$

c $x = -\frac{9}{4}$

d $y = -\frac{9}{8}$

e



43

a

i $(8,7)$

ii $(0,263)$

b

i $(-6,4)$

ii $(0,-140)$

44

a $(0,-12)$

b No

c $x<0$

45

a $x=-3.1$

b -16.61

46

a $x=2,6$

b $y=-24$

c $x=4$

d $y=8$

47

a $x=1.55$

b $y=2.95$

c Yes

d $x=3.58,-0.47$

48

a $x=2.75$

b $y=-9.10$

c Yes

d $x=-0.42,5.91$